Semantic Drift in French Political Discourse: Diachronic Analysis of Legislative Campaign Manifestos (1973–1993) via Temporal Word Embeddings and Topic Modeling

Saimane Nawal
Ensae Paris
nawal.saimane@ensae.fr

Abstract

We analyze the semantic evolution of key political concepts in French legislative campaign manifestos between 1973 and 1993, using the Archelec corpus (CEVIPOF, Sciences Po) comprising 21,697 documents. Our pipeline combines SGNS word2vec models trained per time slice with sequential initialization [Kim et al., 2014] and orthogonal Procrustes alignment [Hamilton et al., 2016], alongside a global LDA topic model to measure the thematic specialization of political families. Results show marked semantic drift for *immigration* ($d_{\cos} = 0.60$), *sécurité* (security, 0.45) and *nation* (0.44). These three concepts converge in LDA toward a single topic over-represented $\times 4.6$ among far-right candidates, reflecting a coherent discursive recomposition that begins around 1978 and intensifies in the late 1980s, coinciding with the FN's electoral breakthrough in 1986.

1 Introduction

Political language is not static: words are ideological battlegrounds whose meaning shifts in response to historical events [Garg et al., 2018]. Distributed word embeddings [Mikolov et al., 2013] now make it possible to *quantify* these semantic shifts from large corpora, opening new avenues for computational political science.

The Archelec corpus (CEVIPOF, Sciences Po) collects campaign manifestos (*professions de foi*) from French legislative election candidates since 1958. Restricted to the five elections from 1973 to 1993, it covers a politically dense period: the Common Programme of the Left (1973), the PS-PCF split (1978), the historic left-wing alternation under Mitterrand (1981), cohabitation (1988), and the conservative landslide (1993). We address the following question:

> *How does the meaning of major political concepts evolve between 1973 and 1993 in French legislative campaign manifestos, and is this evolution driven differently across political families?*

2 Data

2.1 Text corpus

Texts are extracted as OCR transcriptions from the GitLab repository `ckermorvant/arkindex_archelec` (Teklia). Table 1 summarizes the corpus after preprocessing.

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

Table 1: Corpus statistics per election after spaCy preprocessing. Token counts decrease after 1981, likely reflecting an editorial shift toward shorter, more visual manifestos.

Election	Documents	Raw tokens	Lemma tokens	Vocab.	Avg. tokens/doc
1973	3,917	3,114,500	1,157,735	43,330	295
1978	5,025	4,075,666	1,493,092	51,050	297
1981	3,179	2,093,116	768,240	34,990	242
1988	3,615	2,002,003	733,873	34,153	203
1993	5,929	3,881,920	1,459,774	50,378	246
Total	**21,665**	**15,167,205**	**5,612,714**	—	—

2.2 Metadata and political families

The `archelec.csv` file provides 33,031 entries with 42 columns (sex, age, occupation, party affiliation, department, etc.). The join with the text corpus succeeds for 97.6% of documents. The `titulaire-soutien` field is manually grouped into six political families consistent across the period: **far_left** (PCF, LCR, PSU), **left** (PS, MRG), **ecologists** (Greens), **centre_right** (UDF, MRP), **gaullist_right** (RPR, UDR), and **far_right** (Front National). This yields 14,081 labeled documents (65%). Notably, the FN grows from fewer than 150 documents before 1988 to 651 in 1993, this major increase over the period mirrors the party's electoral trajectory. Ecologists surge from 73 documents in 1978 to 1,138 in 1993, driven by the proliferation of Green lists following the Maastricht referendum.

2.3 Keywords Evolution

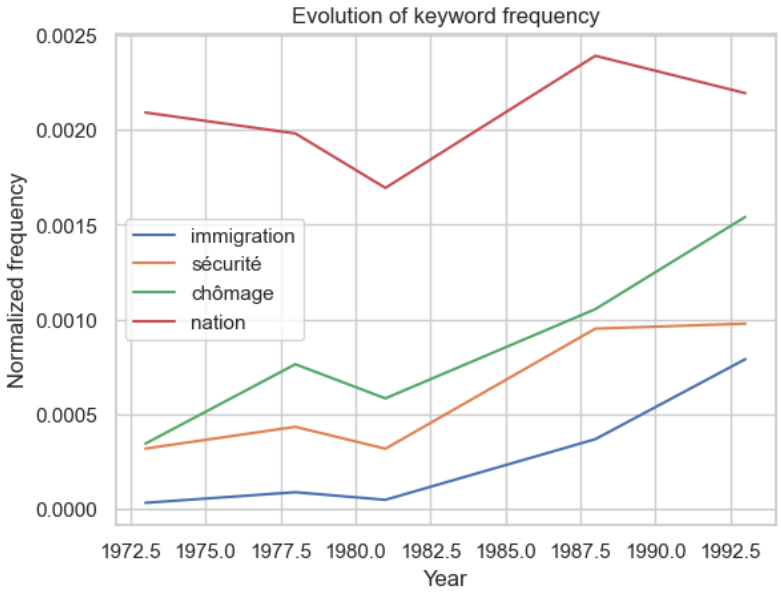

Figure 1: Evolution of normalized frequency for key political terms across elections.

To complement the descriptive statistics, we examine the evolution of the frequency of key political terms. Figure 1 shows that words such as *immigration*, *sécurité*, and *chômage* increase in frequency over time, suggesting a growing prominence of these themes in political discourse.

2.4 Preprocessing

Lemmatization uses `spaCy` (`fr_core_news_lg`). We apply a `has_vector` filter to remove OCR fragments (tokens absent from the language model), a custom stopword list of ∼30 genre-specific

tokens, and set `min_count`$= 15$ in word2vec. End-of-line word splits are regularized via regex before lemmatization.

3 Related Work

Mikolov et al. [2013] introduced word2vec (SGNS). Kim et al. [2014] proposed initializing vectors at time t with those from $t-1$ to reduce stochastic variance and enable temporal comparability. Hamilton et al. [2016] formalized orthogonal Procrustes alignment on the shared vocabulary and identified statistical laws of semantic change over two centuries of English. Garg et al. [2018] applied this approach to quantify gender and ethnic stereotypes over 100 years of American history, demonstrating the method's utility for social science. Kulkarni et al. [2015] formalized statistical significance testing for linguistic change via permutation tests. On moderate-size corpora, static approaches remain competitive with contextual methods [Kutuzov et al., 2018]. For topic modeling, LDA [Blei et al., 2003] remains the dominant approach; thematic specialization of subgroups can be measured via lift indices [Roberts et al., 2014].

4 Methods

4.1 Overall pipeline

The pipeline has two complementary phases: (1) per-period word2vec models with Procrustes alignment to measure diachronic semantic drift; (2) a global LDA model with per-family analysis to identify structuring themes and their evolution.

4.2 Word2Vec and Procrustes alignment

We train one SGNS model per election (hyperparameters: $d = 100$, window$= 8$, `min_count`$= 15$, negative$= 10$, 30 epochs, fixed seed). Initialization follows Kim et al. [2014]: vectors from $t-1$ are copied into model t before training. The initialization ensures that the five embedding spaces remain comparable across periods: without it, each model would learn an arbitrarily oriented vector space, making direct cosine comparisons between years meaningless. Table 2 details inter-period transmission rates, which are high (78–94%), confirming vocabulary stability across elections.

Table 2: Word2Vec vocabulary size and initialization rate from $t-1$.

Year	Vocab.	Init. from $t-1$	% transmitted
1973	5,188	—	—
1978	5,836	4,521	78%
1981	4,076	3,812	94%
1988	3,776	3,214	85%
1993	5,496	3,298	60%

Orthogonal Procrustes alignment [Hamilton et al., 2016] then aligns each model $t \in \{1978, 1981, 1988, 1993\}$ onto the 1973 reference space via rotation R^* minimizing Frobenius distance:

$$R^* = \arg\min_{R^\top R = I} \|A - BR\|_F = VU^\top \quad \text{with} \quad U\Sigma V^\top = A^\top B$$

4.3 Drift metrics

For each word w in the shared vocabulary ($|V_\cap| \approx 2{,}050$ words), we compute **cosine drift** $d_{\cos}(w,t) = 1 - \mathbf{v}_w^{(\text{ref})} \cdot \mathbf{v}_w^{(t)}$ and **Jaccard drift** on the top-10 nearest neighbors. **Mean drift** is averaged over $\{1978, 1981, 1988, 1993\}$.

4.4 LDA topic model

A global LDA model is trained on the union of five corpora with $K = 12$ topics, selected by maximizing C_v coherence over $\{8, 10, 12, 15, 20\}$ ($C_v = 0.483$ at $K = 12$). The **specialization**

index (lift) of family f on topic t in year y is:

$$\text{lift}(f, t, y) = \frac{\bar{\theta}_{f,y,t}}{\bar{\theta}_{y,t}}$$

where $\bar{\theta}_{f,y,t}$ is the mean topic prevalence for documents of family f, and $\bar{\theta}_{y,t}$ is the corpus-wide mean. A lift of 1.0 means the family uses the topic as much as average; a lift of 4.6 means it uses it 4.6 times more than the average candidate.

5 Results

5.1 Semantic drift trajectories

Table 3 shows cumulative cosine drift for eight target terms, and Figure 2 visualizes their trajectories.

Table 3: Cumulative semantic drift of target words (reference: 1973, Procrustes-aligned spaces).

Word	d_{1978}	d_{1981}	d_{1988}	d_{1993}	**Mean drift**
immigration	0.61	0.60	0.62	0.58	**0.60**
sécurité	0.42	0.43	0.46	0.51	**0.45**
chômage	0.43	0.41	0.45	0.46	**0.44**
nation	0.39	0.42	0.45	0.65	**0.44**
travail	0.31	0.37	0.45	0.48	**0.38**
europe	0.39	0.31	0.39	0.36	**0.37**
social	0.27	0.32	0.34	0.46	**0.36**
liberté	0.30	0.31	0.32	0.35	**0.31**

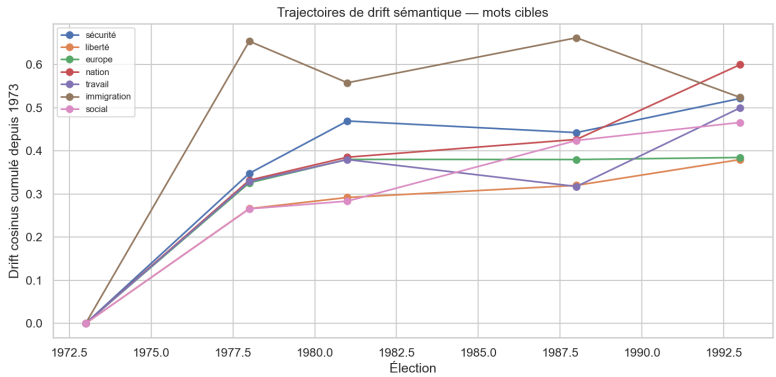

Figure 2: Cumulative cosine drift from 1973. Each point represents one election; a higher value means the word's semantic neighborhood has moved further from its 1973 position.

Immigration shows the highest and earliest drift ($d = 0.61$ already by 1978), which coincides with the PS-PCF split and the rise of economic hardship after the 1973 oil shock, this corresponds to a period during which immigration increasingly became a contested political issue, rather than being primarily framed as a labor market concept. *Sécurité* drifts steadily and accelerates after 1988 ($d = 0.51$ in 1993), tracking the FN's growing electoral presence and its systematic articulation of security with immigration. *Nation* stands out for its sharp 1993 jump ($d_{1993} = 0.65$, the highest single-period value), likely related to the Maastricht Treaty debate: sovereignty and national identity became intensely contested in the run-up to the September 1992 referendum. *Liberté* and *social*, by contrast, remain the most stable words across the period ($d \approx 0.31$–0.36), suggesting they function as consensual political vocabulary across all families.

5.2 Framing shift: *immigration*

To measure whether the type of discourse around *immigration* changed, we compute for each election the mean cosine similarity between the *immigration* vector and two competing lexical fields: an economic frame (*travailleur, ouvrier, emploi, salaire, industrie*) and an identity frame (*national, français, frontière, délinquance, sécurité, criminel*). A higher similarity score means the word is used in contexts closer to that field. In 1973, *immigration* sits closer to the identity frame (0.32) than the economic frame (0.18), but the gap between the two is moderate (+0.14). By 1993, the identity frame has strengthened (0.39) while the economic frame has slightly weakened (0.16), widening the gap to +0.23. In concrete terms: in 1973, *immigration* was still discussed alongside labor and workforce vocabulary; by 1993, its semantic neighborhood was almost exclusively made up of national identity and public order terms. The word appears not only to have drifted semantically, but also to have shifted toward a different dominant register.

5.3 Diachronic neighborhood of *sécurité*

Table 4 provides a granular view of this reframing. In 1973, the five nearest neighbors of *sécurité* (*hygiène, licenciement, garantir, procurer, égalité*) all belong to the welfare state register, the *sécurité* referred to here is *sécurité sociale* (social security). By 1993, the neighborhood has completely turned over: *police, peine_mort, délinquance, immigration* now define the word's meaning. The transition is gradual: *bandit* and *ordre* appear as early as 1978, signaling the beginning of the semantic shift well before the FN's mainstream breakthrough.

Table 4: Top-5 cosine neighbors of *sécurité* per election (Procrustes-aligned spaces). The shift from social welfare to law-and-order vocabulary appears to be largely completed by 1993.

Rank	1973	1978	1981	1988	1993
1	hygiène	bandit	bandit	culture	renforcer
2	licenciement	santé	ordre	garantir	police
3	garantir	liberté	défense	solidarité	peine_mort
4	procurer	confort	citoyen	environnement	délinquance
5	égalité	ordre	paix	tolérance	**immigration**

5.4 Topic modeling and partisan specialization

The 12 LDA topics are politically interpretable. The most relevant are T1 (*national, front, force, député*), T5 (*national, immigration, france, impôt, chômage, rétablir*) and T8 (*écologie, environnement, solidarité, développement*).

The **lift** measures how much more or less a political family uses a given topic compared to the average candidate. A lift of 1.0 is baseline; a lift of 4.6 means a family is 4.6 times more likely to write about that topic than a random candidate. Table 5 crosses drift results with topic specialization.

Table 5: Semantic drift $\times$ dominant LDA topic $\times$ political family. The four highest-drift words all belong to the same topic, which the far right uses 4.6$\times$ more than average.

Word	Mean drift	Dominant LDA topic	Family	Max lift
immigration	0.60	T5 (national, immigration, france)	far_right	4.62
sécurité	0.45	T5 (idem)	far_right	4.62
chômage	0.44	T5 (idem)	far_right	4.62
nation	0.44	T5 (idem)	far_right	4.62
travail	0.38	T6 (travailleur, droit)	ecologists	4.11
europe	0.37	T8 (écologie, environnement)	ecologists	7.84
social	0.36	T0 (social, entreprise, liberté)	centre_right	2.00
liberté	0.31	T0 (idem)	centre_right	2.00

The four words with the highest drift (*immigration, sécurité, chômage, nation*) all converge on topic T5, which the far right uses 4.6 times more than the average candidate. For context: other families

show lifts between 1.2 and 2.0 on their most distinctive topics, making the far right's $\times 4.6$ an outlier. Ecologists show the second-most distinctive specialization ($\times 7.84$ on T8 for *europe*, $\times 10.8$ on T8 in 1988). Unlike the far right, whose distinctive topic simultaneously captures four high-drift concepts, the ecologists' specialization is concentrated on a single thematic domain: **environmental policy**, without reorganizing the broader semantic field around it.

6 Discussion

Between 1973 and 1993, four political concepts: *immigration*, *sécurité*, *chômage*, and *nation*; underwent the most significant semantic transformations in the corpus, and they did so jointly. In 1973, these words lived in separate semantic neighborhoods: *sécurité* meant social protection, *chômage* described a structural economic condition, *immigration* referred to labor flows, and *nation* carried a broad civic resonance. By 1993, all four had migrated toward the same discursive space, forming a unified topic (T5) in which unemployment, immigration, national identity, and public order are presented as a single interconnected problem.

This recomposition was not gradual and uniform. The drift trajectories reveal a two-phase dynamic: a first shift beginning in 1978 (the year of the PS-PCF split and a period marked by the consolidation of rising unemployment following the mid-1970s economic crisis) and a sharper acceleration after 1988, coinciding with the FN's consolidation as a major electoral force. The neighborhood analysis of *sécurité* makes this timeline concrete: *bandit* and *ordre* appear as early as 1978, while *police*, *délinquance*, and *immigration* only dominate by 1993. The semantic change did not happen overnight; it was built incrementally across three electoral cycles.

Crucially, this recomposition is asymmetric across political families. Topic T5 is strongly over-represented among far-right candidates (×4.6 relative to the corpus average), while other families remain near baseline on this topic. The other families show their own distinctive signatures. For instance, ecologists on environmental themes, or the centre-right on social-liberal vocabulary. However, none show the same degree of cross-concept coupling. This suggests that what we are measuring is not a general societal shift in how these words were used, but a specific discursive strategy: a consistent co-occurrence and semantic association between immigration, security, unemployment, and national belonging. The far right did not simply talk more about immigration; it reorganized the semantic field around it.

Limitations. (i) *OCR quality*: residual artefacts persist despite `has_vector` filtering; topic T10 mixes Alsatian regionalism with unfiltered German OCR tokens. (ii) *Partial metadata*: 35% of documents lack a political family label, biasing analyses toward structured parties. (iii) *Shared vocabulary constraint*: Procrustes alignment relies on the $\sim$2,050 words present in all five periods, mechanically excluding lexical innovations such as *maastricht* or disappearances such as *programme commun* [Hamilton et al., 2016]. (iv) *LDA bag-of-words*: syntactic structure is ignored; a whitelist of 33 political bigrams partially mitigates this.

7 Conclusion

From 21,697 legislative manifestos spanning two decades of French political history, we have shown that the semantic recomposition of *immigration*, *sécurité*, *chômage*, and *nation* into a single discursive complex is both quantitatively measurable and historically interpretable. This result is obtained without any supervised signal about political ideology, relying solely on distributional patterns in the text. The method is conservative: only the four highest-drift words converge on the same topic, and the far right's $\times 4.6$ specialization on that topic stands as a robust, unsupervised finding.

Go further: Three extensions are natural: (i) a Structural Topic Model integrating political family as a covariate [Roberts et al., 2014], which would allow controlling for confounds such as department or candidate profession; (ii) contextual embeddings (CamemBERT, Martin et al. 2020) for richer framing analysis; (iii) cross-referencing with departmental economic data to test whether topic T5 prevalence correlates with local unemployment rates.

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